GM UNIVERSITY DAVANAGERE | DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

HARDWARE WIRING & CIRCUIT INTERCONNECT MANUAL

Course: UE25EC2302 (Hardware Modelling using Verilog) | Project: Railway Level Crossing Controller

Semester: 3 | **Marks:** 40 PBL

HARDWARE BENCH SPECIFICATION

Vikas K S (Lead): U26E02EC521 (Arch/FSM) **Nandan S L:** U26E02EC518 (Safety Interlock) **Meghana P:** U25E02EC070 (Traffic/ Actuators) **Keerthana R F:** U25E02EC053 (Verification/Synth)

1. POWER DISTRIBUTION NETWORK & DECOUPLING ARCHITECTURE

CRITICAL HARDWARE RULE (COMMON GROUND & BROWNOUT PREVENT): The two SG90 barrier servos draw up to **1.6A peak current** during initial rotational acceleration. They must be powered from a dedicated **External 5.0V 2.0A DC Power Supply**. The Ground (GND) of the external supply and the Arduino/FPGA GND **MUST BE TIED TOGETHER (COMMON GROUND)**. A **470 µF electrolytic bulk capacitor** in parallel with a **0.1 µF ceramic bypass disc** across the motor rail suppresses inductive voltage droops.

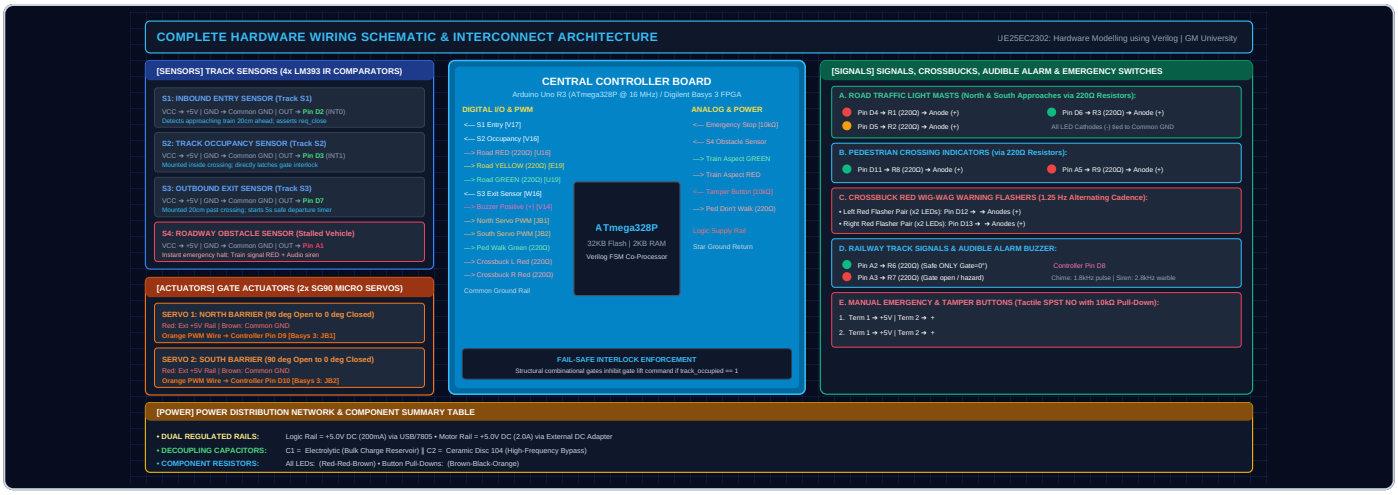
Power Rail	Voltage & Source	Current Cap	Filter Network	Connected Load Subsystems
Logic VCC (+5V)	+5.0V DC (USB / 7805 Reg)	200 mA max	0.1 µF Ceramic on VCC	ATmega328P, 4x LM393 IR Sensors, Buttons
Motor VCC (+5V_EXT)	+5.0V DC (External Adapter)	2000 mA (2.0A)	**470 µF / 16V ‖ 0.1 µF / 50V**	TowerPro SG90 Servos (North & South)
Common GND (0V)	Common Star Return Rail	Return Path (0V)	Single Low-Z Ground Bus	MCU GND, Ext Supply GND, All LED Cathodes

2. MASTER PIN-TO-PIN INTERCONNECT & NETLIST MATRIX

Sl.	Component Name	Device Terminal	Arduino Uno Pin	Basys 3 FPGA	Series Element	Signal Net & Function
1	**Entry Sensor (S1)**	Digital OUT (Pin 3)	**Pin D2 (INT0)**	V17 (SW0)	Internal Pull-up	`sensor_inbound` (Active-LOW)
2	**Track Occupancy (S2)**	Digital OUT (Pin 3)	**Pin D3 (INT1)**	V16 (SW1)	Internal Pull-up	`track_occupied` (Active-LOW)
3	**Exit Sensor (S3)**	Digital OUT (Pin 3)	**Pin D7**	W16 (SW2)	Internal Pull-up	`sensor_outbound` (Active-LOW)
4	**Obstacle Sensor (S4)**	Digital OUT (Pin 3)	**Pin A1**	W17 (SW3)	Internal Pull-up	`obstacle_detected` (Emergency)
5	**Manual Emergency Btn**	NO Terminal 2	**Pin A0**	U18 (BtnC)	**10 kΩ Pull-Down**	`emergency_btn` (Active-HIGH)
6	**Gate Tamper Button**	NO Terminal 2	**Pin A4**	T18 (BtnU)	**10 kΩ Pull-Down**	`tamper_test` (Interlock verify)
7	**North Barrier Servo**	Orange Lead (PWM)	**Pin D9 (PWM)**	JB1 (J1)	Direct PWM	`servo_north_pwm` (50 Hz pulse)
8	**South Barrier Servo**	Orange Lead (PWM)	**Pin D10 (PWM)**	JB2 (L2)	Direct PWM	`servo_south_pwm` (50 Hz pulse)
9	**Road RED Light (Stop)**	Anode (+) Lead	**Pin D4**	U16 (LD0)	**220 Ω, 1/4W (R1)**	`road_red` (Halt road traffic)
10	**Road YELLOW (Warn)**	Anode (+) Lead	**Pin D5 (PWM)**	E19 (LD1)	**220 Ω, 1/4W (R2)**	`road_yellow` (Pre-closure warning)
11	**Road GREEN (Go)**	Anode (+) Lead	**Pin D6 (PWM)**	U19 (LD2)	**220 Ω, 1/4W (R3)**	`road_green` (Safe 90° open)
12	**Pedestrian WALK**	Anode (+) Lead	**Pin D11**	V19 (LD3)	**220 Ω, 1/4W (R8)**	`ped_walk_green` (Pedestrian cross)
13	**Pedestrian DON'T WALK**	Anode (+) Lead	**Pin A5**	U15 (LD5)	**220 Ω, 1/4W (R9)**	`ped_stop_red` (Pedestrian halt)
14	**Crossbuck Flasher L**	Anodes (x2 LEDs)	**Pin D12**	W18 (LD4)	**220 Ω, 1/4W (R4)**	`crossbuck_l` (1.25 Hz Wig-Wag)
15	**Crossbuck Flasher R**	Anodes (x2 LEDs)	**Pin D13**	U14 (LD6)	**220 Ω, 1/4W (R5)**	`crossbuck_r` (180° Inverted phase)
16	**Train GREEN Aspect**	Anode (+) Lead	**Pin A2**	V14 (LD7)	**220 Ω, 1/4W (R6)**	`train_green` (Gate = 0° safe)
17	**Train RED Aspect**	Anode (+) Lead	**Pin A3**	V13 (LD8)	**220 Ω, 1/4W (R7)**	`train_red` (Gate open / obstacle)
18	**Active Piezo Buzzer**	Positive (+) Pin	**Pin D8**	W19 (PMOD)	Direct Drive	`buzzer_pwm` (1.8kHz/ 2.8kHz tone)

3. MASTER ELECTRICAL WIRING & INTERCONNECT SCHEMATIC

The complete system interconnect schematic illustrates all sensor conditioning inputs, servo PWM drives, current-limited signal masts, audible transducer, emergency switches, and decoupling filter network:



4. SUBSYSTEM ELECTRICAL SCHEMATICS & INTERFACE CALCULATIONS

A. Track Proximity Sensors (4x LM393)

Principle: 940 nm GaAs emitter + phototransistor. On-board LM393 comparator gives clean active-LOW digital logic.

- **VCC / GND:** Connected to +5V Logic Rail and Common GND.
- **D0 Digital Output:** Pulls to **0.0V (LOW)** on train detection; sits at **5.0V (HIGH)** in clear quiescent state.
- **Range Tuning:** Rotate 10kΩ trimpot clockwise to increase distance (ideal: 3 cm to 5 cm from track centerline).

B. Barrier Servos (2x TowerPro SG90)

Principle: 50 Hz PWM positional control (Period T = 20.0 ms). Pulse width dictates rotational angle.

- **Vertical Position (Open 90°):** 2.0 ms high pulse (50 Hz frame).
- **Horizontal Position (Closed 0°):** 1.0 ms high pulse (50 Hz frame).
- **Power Rail:** Red wire to External 5V 2A; Brown wire to GND.
- **Ramp Rate:** ±1° step every 27.7 ms eliminates gear stress.

C. Traffic & Aspect LEDs (Current-Limiting)

Ohm's Law Calculation: R = (V_CC - V_F) / I_F = (5.0V - 2.0V) / 0.015A = 200 Ω → **220 Ω ±5%, 1/4W**

- **Resistor Rating:** 220 Ω, 1/4W Metal Film (Red-Red-Brown-Gold).
- **Safe Current:** I_F = 13.6 mA per pin (well within 20mA limit).
- **Wiring:** Resistor in series with Anode (+); Cathodes (-) to GND.

D. Push Buttons & Audible Buzzer

Pull-Down Resistor Calculation: I_closed = 5.0V / 10 kΩ = 0.5 mA (Negligible power loss)

- **Resistor:** 10 kΩ, 1/4W (Brown-Black-Orange-Gold) ties pin to 0.0V.
- **Action:** Pressing button injects solid 5.0V pulse without floating.
- **Active Buzzer:** Pin D8 (+) to GND (-); 1.8kHz chime / 2.8kHz siren.

5. BENCH ASSEMBLY & DIORAMA CONSTRUCTION PROCEDURE

Step 1: Breadboard Power Bus Configuration

1. Place the 830-point breadboard horizontally on your workbench. Connect the upper red power rail to the **External 5V 2A DC Supply (+)** and upper blue rail to **External GND (-)**.
2. Connect the lower red rail to the **Arduino +5V pin** and lower blue rail to the **Arduino GND pin**.
3. **TIE COMMON GROUND:** Insert a heavy jumper wire connecting the upper blue rail to the lower blue rail. Both power supplies now share the same 0.0V reference potential.
4. Insert the **470 µF electrolytic capacitor** across the upper rail (Long lead = +5V, Stripe lead = GND). Insert the **0.1 µF ceramic disc capacitor (104)** in parallel.

Step 2: Track Sensor Alignment & Potentiometer Calibration

1. Mount **Sensor 1 (Entry)** 20 cm before the crossing on the left side of the track. Angle the IR transmitter/receiver towards the rail track at 45°.
2. Mount **Sensor 2 (Occupancy)** directly in the center of the level crossing diamond facing across the rails.
3. Mount **Sensor 3 (Exit)** 20 cm past the crossing on the departure side.
4. Mount **Sensor 4 (Obstacle)** facing the roadway intersection to detect stalled vehicles.
5. **Calibration Routine:** Power up the system. Place a toy train or wagon on the track in front of Sensor 1. Using a precision screwdriver, turn the onboard brass trimpot until the green D0 indicator LED on the sensor module lights up. Move the train away: the LED must turn off immediately. Repeat for Sensors 2, 3, and 4.

Step 3: Motorized Barrier Horn Mechanical Zeroing

1. Connect Servo 1 signal to Pin D9 and Servo 2 signal to Pin D10.
2. Power on the Arduino with the companion driver sketch. The firmware automatically sends a 90° (vertical) command to both servos upon boot.
3. Press the single-arm servo horn onto the servo spline such that the barrier arm stands **vertically upright at exactly 90°**. Tighten the central horn screw.
4. Trigger Sensor 1: verify that both barrier arms rotate downward smoothly without grinding or stalling until they rest **horizontally across the roadway at 0°**.

6. HARDWARE TROUBLESHOOTING & DIAGNOSTIC MATRIX

Observed Symptom	Probable Root Cause	Bench Measurement / Check	Corrective Action
Servo Jitter / Board Reset	Current brownout from Arduino 5V regulator	Measure 5V rail with multimeter during motor sweep (drops below 4.2V)	Power servos from external 5V 2A supply with 470µF bulk capacitor
Erratic / Uncontrolled Servo Motion	Missing Common Ground connection	Measure DC voltage between Arduino GND and Power Adapter GND	Connect jumper wire between Arduino GND and external supply GND
IR Sensor Always ON (D0=0V)	Ambient sunlight or trimmer sensitivity too high	Observe module status LED under ambient lighting	Turn potentiometer counter-clockwise to reduce optical threshold
Emergency Button Trips Spontaneously	Floating CMOS input (missing pull-down)	Measure voltage on Pin A0 when button is unpressed (floats at ~1.8V)	Add 10 kΩ pull-down resistor from Pin A0 to Common GND
Dim / Non-illuminating LEDs	Reversed LED polarity or incorrect resistor	Check long lead (Anode +) vs flat edge (Cathode -)	Verify 220 Ω resistor value (Red-Red-Brown); reverse LED orientation

7. COMPANION ARDUINO HARDWARE DRIVER SKETCH (MATCHING VERILOG RTL FSM)

Replicates the exact 4-state Moore FSM and hardware safety interlocking modeled in course **UE25EC2302**:

```
#include <Servo.h>
const int PIN_S1 = 2, PIN_S2 = 3, PIN_RED = 4, PIN_YEL = 5, PIN_GRN = 6, PIN_S3 = 7, PIN_BUZZ = 8;
const int PIN_SRV_N = 9, PIN_SRV_S = 10, PIN_PED_W = 11, PIN_CR_L = 12, PIN_CR_R = 13;
const int PIN_EMG = A0, PIN_OBST = A1, PIN_TRN_G = A2, PIN_TRN_R = A3, PIN_PED_S = A5;

enum FSMState { ST_OPEN = 0, ST_CLOSING = 1, ST_CLOSED = 2, ST_OPENING = 3 };
FSMState state = ST_OPEN;
Servo srvN, srvS;
int angle = 90;

void setup() {
  pinMode(PIN_S1, INPUT); pinMode(PIN_S2, INPUT); pinMode(PIN_S3, INPUT); pinMode(PIN_OBST, INPUT); pinMode(PIN_EMG, INPUT);
  pinMode(PIN_RED, OUTPUT); pinMode(PIN_YEL, OUTPUT); pinMode(PIN_GRN, OUTPUT); pinMode(PIN_BUZZ, OUTPUT);
  pinMode(PIN_PED_W, OUTPUT); pinMode(PIN_PED_S, OUTPUT); pinMode(PIN_CR_L, OUTPUT); pinMode(PIN_CR_R, OUTPUT);
  pinMode(PIN_TRN_G, OUTPUT); pinMode(PIN_TRN_R, OUTPUT);
  srvN.attach(PIN_SRV_N); srvS.attach(PIN_SRV_S);
  srvN.write(90); srvS.write(90);
}

void loop() {
  bool inb = (digitalRead(PIN_S1) == LOW), occ = (digitalRead(PIN_S2) == LOW);
  bool out = (digitalRead(PIN_S3) == LOW), obs = (digitalRead(PIN_S4) == LOW);
  bool emg = (digitalRead(PIN_EMG) == HIGH), hazard = (inb || occ || obs || emg);

  switch (state) {
    case ST_OPEN:    if (hazard) state = ST_CLOSING; break;
    case ST_CLOSING: if (angle > 0) { angle--; delay(25); } else state = ST_CLOSED; break;
    case ST_CLOSED:  if (!hazard && out) state = ST_OPENING; break;
    case ST_OPENING: if (hazard) state = ST_CLOSING; else if (angle < 90) { angle++; delay(25); } else state = ST_OPEN; break;
  }
  srvN.write(angle); srvS.write(angle);

  digitalWrite(PIN_GRN, (state == ST_OPEN && !hazard));
  digitalWrite(PIN_YEL, (state == ST_CLOSING));
  digitalWrite(PIN_RED, (state == ST_CLOSED || state == ST_CLOSING || hazard));
  digitalWrite(PIN_TRN_G, (state == ST_CLOSED && angle == 0 && !obs && !emg));
  digitalWrite(PIN_TRN_R, !(state == ST_CLOSED && angle == 0 && !obs && !emg));
  digitalWrite(PIN_PED_W, (state == ST_OPEN && !hazard));
  digitalWrite(PIN_PED_S, !(state == ST_OPEN && !hazard));

  if (state == ST_CLOSING || state == ST_CLOSED || hazard) {
    bool blink = (millis() / 400) % 2;
    digitalWrite(PIN_CR_L, blink); digitalWrite(PIN_CR_R, !blink);
    digitalWrite(PIN_BUZZ, (obs || emg) ? HIGH : blink);
  } else {
    digitalWrite(PIN_CR_L, LOW); digitalWrite(PIN_CR_R, LOW); digitalWrite(PIN_BUZZ, LOW);
  }
}
```

8. DEMONSTRATION PROTOCOL FOR EVALUATION (40 MARKS PBL)

1. **Baseline Safe Check:** Power ON. Road Green ON, Pedestrian Walk ON, Barrier at 90° upright, Train Signal RED.
2. **Train Arrival Demonstration:** Slide toy train past Sensor 1. Observe Road Yellow ON, Buzzer chimes, Crossbuck flashers alternate (1.25 Hz), Barrier descends smoothly from 90° to 0°.
3. **Interlock Lock Confirmation:** When barrier reaches 0°, observe Train Aspect switch to **GREEN (PROCEED)**.
4. **Departure Clearance:** Move train past Sensor 3. 5-second buffer passes, barrier smoothly rises back to 90°, Road Green restored.
5. **Emergency Obstacle Injection:** Place car on tracks during green signal. Sensor 4 immediately trips Train Signal RED and activates continuous emergency siren.

— End of Official Hardware Wiring & Circuit Interconnect Manual (UE25EC2302) —